

Software Requirements Specification

BusBuddy -

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Field	This example
Team	Team25
Project title	FindTutor
SDLC chosen in M1 + reason	Agile - A flexible framework that breaks down the project into smaller, much manageable cycles
Date / version	v0.1
GitHub repo	n/a

1. Introduction

1.1 Purpose and scope

Students will sometimes find themselves in need of a tutor. Finding a tutor can be quite difficult, especially one who can effectively teach the course subjects as well as having schedules that match theirs. On the other hand, some students may want to help other students but they have difficulty finding ones that require their expertise. FindTutor is a tutoring platform which allows students to either sign up to be a tutor, list the courses that they can tutor and when they are available for tutoring, or they can look for tutors based on their desired courses and availability, contact the tutors, and leave review for them.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Students	primary user	<u>(Tutors)</u> Being able to offer tutoring and manage availability <u>(Learners)</u> Find and book tutors	Leaks of private information, and fake reviews.
Course Instructors	secondary stakeholder	Students being able to receive academic support	Tutors feeding incorrect information and illegal misconduct being performed in the app.
University	institution supplier or	A secure and reliable that is easy to maintain	Private data being leaked, security vulnerabilities, and data breach

2. Overall description

2.1 Product context — what it does / does not

FindTutor can find

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-1 List what courses they can tutor and available time to tutor	• The user's safety if they were to meet in-person • A video call system • Any payment or subscription feature

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-2 Look for tutors with a filter system based on what subject they're looking for • FR-3 View tutor availability • FR-4 Request/receive tutoring requests/Online communication with tutor and student • FR-5 Leave a review	• AI translator

2.2 Assumptions and constraints

- Assumption: Students have access to a device that can be connected to the internet
- Assumption: Students that are looking to be a tutor would be fine with not being paid in a way
- Constraint: Thai and English UI text
- Constraint: The user's safety - If both of the students were to meet in-person, we can not in any way guarantee their safety if one of them was malicious.
- Constraint: 10-week build window shared with four other milestones (M1–M5).

3. Functional requirements

FR-1 — List what courses they can tutor and available time to tutor

User story: As a student, I want to tutor fellow students who may need my expertise and for them to know what time I'm available, so that I may be able to tutor them effectively

Pain this traces to: Student tutors can have some difficulty finding a fellow student that requires their help can be challenging, and even if they found one, having mismatched schedules can be even more troublesome.

1. Given courses and the available time, when the student tutor saves it, then the courses they can tutor and available time will appear on their public profile.
2. Given an already existing profile, when the student tutor edits it, then the updated profile is shown immediately and no duplicated profile is created.

FR-2 — Look for tutors with a filter system based on what subject they're looking for

User story: As a student, I want to be able to find a tutor based on the courses I have listed in the filter, so that I can find a tutor that matches my needs.

Pain this traces to: Finding a tutor. can be hard, and it's even harder to find a tutor that can teach in a specific subject.

1. Given the courses the student has inputted, when the student searches for a tutor, then it will show the tutors with the listed courses and hide the ones without
2. Given there are not a single tutor with the matching courses, when the student searches for a tutor, then it will inform the students that no matching tutors were found

FR-3 — View tutor availability

User story: As a student, I want to be to see what time the tutor is available for tutoring, so both the tutor and I will be able to have a tutoring session

Pain this traces to: Tutors and tutees have to exchange messages after meeting each other to know their availability

1. Given the tutor entered their schedule/availability, when the students look at the profile, then, they will be able to see the available time for tutoring
2. Given the tutor is unavailable and the tutor has listed in the profile they're not available, when the students look at the profile, then they will see that the tutor is unavailable.

FR-4 — Request/receive tutoring requests/Online communication with tutor and student

User story: As a student, I want to be able request a tutoring session to the tutor, so that I may be able to have a session with them depending on if they accepted or declined.

Pain this traces to: Students and tutors have to use a separate messaging platform to arrange a tutoring session, which can hard to keep track of, especially if the student or tutor have many notifications filling up their screen

1. Given the student has selected a tutor with the time for the session e.g. (10:30 - 12:25), when the student submits the request to tutor profile, then the tutor will be notified and will be able to accept or reject the request
2. Given the student has selected the same tutor for request and they have yet to accept or reject the request,, when the student send the request, then the platform will notify that the student has already sent the request and will not send the request
3. Given the tutor has received the request, when they accept the request, then the student will see that they have accepted the request and will be able to start chatting with the tutor online
4. Given the tutor has received the request, when they reject the request, then the student will be notified that the request has been rejected and will not be able to send another request for the next 24 hours

FR-5 — Leave a review

User story: As a student, I want leave a review for the tutor, so other students can make informed decisions based on that review

Pain this traces to: Students have limited information about the quality and qualifications of the tutor before they meet

1. Given the student has finished the session whether on time or early, when they submit a review and rating, then the review will appear publicly on the tutor's profile
2. Given the student has already submitted a review, when they edit a review and resubmit it, then the new review will appear publicly on the tutor's profile and no duplicate review will be created

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-5 (security/privacy) is required for M2 — every SRS must have at least one.

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	When a student searches for a tutor, the search results will appear within 5 seconds on campus Wi-Fi	Stopwatch on a device connected to campus Wi-Fi, 3 trials
NFR-2	performance	A tutoring request will be notified to tutor within 5 seconds on campus Wi-Fi	Stopwatch with two devices connected to campus Wi-Fi, 3 trials
NFR-3	usability	A new user should be able to search for a tutor based on a course in less than 3 steps upon opening the app	Give the app to a student who has never seen the app and tell to search for a tutor with <i>insert subject</i> and count the steps they take to find one
NFR-4	reliability	No duplicate profile or review is created when you try to create a new profile or review	Create a profile or a review, create another one, check for any duplicates
NFR-5	security / privacy (required)	When accessing a profile, user must authenticate with their password or id	Check for if the password or id matches, if not, deny access

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Create a tutor profile	Student tutor	FR-1	User creates a tutor profile with their available schedule and what courses they tutor, it will be seen publicly
UC-2 Looking for tutor who can teach 'Intensive English'	Student	FR-2	User chooses and enters in 'Intensive.English' in the search filter, and it shows the tutors with 'Intensive English' listed in their profile
UC-3 Show the student tutor is unavailable to the student	Student tutor → Student	FR-3	The student tutor list in their profile that they're unavailable, when the students look at their profile, they can see the tutor's schedule but the tutor is currently unavailable
UC-4 Send a tutoring request to the tutor	Student → Student tutor	FR-4	The user sends a tutoring request with the time they wish to schedule the tutoring (They cannot enter a time that the tutor is unavailable during) and the tutor will be sent a notification for the tutoring request
UC-5 Leave a 5-star review for the tutor	Student → Student tutor	FR-5	After the session ends based on the chosen time or both the student tutor and student chooses to end it early, the student writes a 5-star review for tutor and can be seen on the tutor profile publicly

5.2 Use-case diagram

Not drawn yet on purpose. Week 3's job is naming these five use cases correctly against the FR list above — the actor/association/<<include>> diagram itself is Week 4's task, taught in the next session.

6. Out of scope

1. The user's safety if they were to meet in-person
2. A video call system
3. Any payment or subscription feature
4. AI translator

7. Traceability matrix (Golden Thread)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Student tutors can have some difficulty finding a fellow student that requires their help can be challenging, and even if they found one, having mismatched schedules can be even more troublesome	FR-1 List what courses they tutor and available time to tutor	(Week 5)	(Week 6-7)	(Week 7)
Finding a tutor. can be hard, and it's even harder to find a tutor	FR-2 Look for tutors with a filter system	(Week 5)	(Week 6-7)	(Week 7)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
that can teach in a specific subject.	based on what subject they're looking for			
Tutors and tutees have to exchange messages after meeting each other to know their availability	FR-3 View tutor availability	(Week 5)	(Week 6–7)	(Week 7)
Students and tutors have to use a separate messaging platform to arrange a tutoring session, which can hard to keep track of, especially if the student or tutor have many notifications filling up their screen	FR-4 Request/receive tutoring requests/Online communication with tutor and student	(Week 5)	(Week 6–7)	(Week 7)
Students have limited information about the quality and qualifications of the tutor before they meet	FR-5 Leave a review	(Week 5)	(Week 6–7)	(Week 7)

8. AI usage log

If this table is empty but the prose looks generated, the milestone is capped at 50% — log every prompt that drafted a story or paragraph, honestly.

Date	Tool model /	What we asked	What AI produced	What we changed / verified	What we learned

9. Self-check

- ✓ Every FR is a user story with a benefit — all 5 FRs in §3 end in a “so that” clause tied to a real pain.
- ✓ Every FR has acceptance criteria you could actually test — 2 AC per FR, each a number, a state, or a yes/no.
- ✓ No FR is vague — no bare “fast”, “easy”, or “user-friendly” anywhere in §3.
- ✓ NFRs exist, including security / privacy — NFR-5 is a required access-control check.
- ✓ Every FR traces back to a stated problem — see “Pain this traces to” under each FR, and §7.
- ✓ Scope and out-of-scope are stated — §2.1 and §6.
- ✓ Use-case list matches the FR list — 5 use cases, one per FR (diagram added Week 4).
- ✓ AI log is honest — §8 shows exactly what was asked, drafted, and hand-corrected.

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.